

MRI-based Habitat Analysis for the Prediction of Progression-Free Survival in Primary Spinal Tumors

Qizheng Wang, MD^{*1} • Yang Zhang, PhD^{*2} • Tongyu Wang, MM³ • Yongye Chen, MD¹ • Ruixin Yan, MM¹ • Ke Liu, MM¹ • Weili Zhao, MM¹ • Dapeng Hao, MD³ • Min-Ying Su, PhD^{**4,5} • Ning Lang, MD^{**1,6}

* Q.W. and Z.Y. contributed equally to this work.

** N.L. and M.Y.S. are co-senior authors.

Author affiliations, funding, and conflicts of interest are listed at the end of this article.

See also the editorial by Jabehdar Maralani and Burgess in this issue.

Radiology 2025; 317(1):e242993 • <https://doi.org/10.1148/radiol.242993> • Content codes:   

Background: Prediction of prognosis in patients with primary spinal tumors is critical for treatment planning but a clinical challenge.

Purpose: To develop an MRI-based nested habitat radiomics model to predict progression-free survival (PFS) in patients with primary spinal tumors.

Materials and Methods: This dual-center retrospective study analyzed patients with resected primary spinal tumors. Patients from March 2010 to August 2019 were assigned to a training set; from September 2019 to October 2022, to an internal test set; and from March 2010 to October 2022, to an external test set. Whole-tumor regions of interest were delineated on T1- and T2-weighted MRI scans. Nested habitat radiomic analysis was performed to locate aggressive subregions associated with a poorer prognosis. Initially, a support vector machine model classified tumors based on global features. Then, local radiomics features were extracted to generate probability maps to identify aggressive tumor subregions with use of *k*-means clustering. Features from tumor microregions with the highest probability were used to build a final habitat model. Performance was evaluated using the area under the receiver operating characteristic curve (AUC) and Kaplan-Meier survival analysis.

Results: Among 259 patients (median age, 39 years [IQR, 28–53 years]; 150 female), the habitat model better predicted PFS compared with the whole-tumor radiomics model (AUC [training set], 0.93 [95% CI: 0.87, 0.98] vs 0.82 [95% CI: 0.68, 0.93]; *P* = .03, DeLong test). The habitat model combined with clinical characteristics showed the best performance (AUC [training set], 0.95 [95% CI: 0.88, 0.99]; AUC [internal test set], 0.86 [95% CI: 0.69, 0.96]; AUC [external test set], 0.89 [95% CI: 0.76, 0.96]). The nested habitat radiomics score was an independent risk factor for 3-year PFS (high- vs low-risk group PFS: 24 vs 28 months; log-rank *P* = .04) in the external test dataset.

Conclusion: In patients with primary spinal tumors, an MRI-based nested habitat radiomics model outperformed other models in predicting PFS.

© RSNA, 2025

Supplemental material is available for this article.

The axial skeleton is a common site for primary and metastatic bone tumors. Primary spinal tumors have a wide range of aggressiveness (1). The World Health Organization classifies them into benign, intermediate locally aggressive non-metastatic, intermediate locally aggressive rarely metastatic, and malignant categories. Surgery is the standard treatment strategy for primary spinal tumors, providing benefits including spinal canal decompression, pain relief, and spinal reconstruction (2). However, the heterogeneity in behavior of the tumors and the complexity of surrounding critical anatomic structures pose substantial challenges for clinicians, who must balance patients' quality of life with minimizing postsurgery recurrence rates (3,4). Accurate prediction of prognosis is critical for individual management.

Despite its importance, predicting postoperative survival remains a considerable clinical challenge with little consensus. Most scoring systems, such as the Tomita, Tokuhashi, Spinal Instability Neoplastic Score, and Kostuik classification systems, have been primarily developed for spinal metastases (5–8). There are few scoring systems available for predicting outcomes in primary spinal tumors. Although a primary spinal tumor mortality score has been proposed previously, the dataset consisted predominantly of benign tumors (9). To our

knowledge, there is no scoring system for patients with aggressive primary spinal tumors.

Imaging plays a critical role in the preoperative evaluation of spinal tumors (10). Various imaging analyses have also been developed for prognostication in patients with cancer by understanding key features extracted from images (11). Artificial intelligence–assisted analysis can also provide imaging markers for predicting prognoses for various cancers and offer opportunities for individualized treatment (12–14). However, the lack of explainability of the artificial intelligence model is a common problem. Recently, habitat analysis has gained wide attention in medical imaging. Unlike conventional radiomics methods that extract features from the whole tumor, this approach divides tumors into subregions called habitats, consisting of voxels with similar attributes likely to be related to tumor biology (15). In a heterogeneous tumor, habitat analysis can localize zones and extract features representing different structures or biologic behavior (16). Compared with deep learning–based analyses, the habitat radiomics method can reduce computational costs and does not require a large training dataset (17). The objective of this study was to develop and evaluate an MRI-based nested habitat radiomics model to predict progression-free survival (PFS) in patients with primary spinal tumors.

Abbreviations

AUC = area under the receiver operating characteristic curve, HR = hazard ratio, PFS = progression-free survival, ROI = region of interest

Summary

In patients with primary spinal tumors, an MRI-based nested habitat clinical-radiomics model outperformed other models in predicting postoperative progression-free survival.

Key Results

- In a retrospective study of 259 patients with primary spinal tumors, MRI-based nested habitat radiomics analysis was applied to locate aggressive subregions in heterogeneous tumors.
- The nested habitat model combined with clinical characteristics showed the best performance in predicting postoperative progression-free survival (PFS) (area under the receiver operating characteristic curve, 0.89) for the external test set.
- The nested habitat radiomics score was an independent risk factor for 3-year PFS (high- vs low-risk group PFS: 24 vs 28 months; log-rank $P = .04$) in the external test set.

Materials and Methods

This retrospective study was approved by the institutional medical science research ethics committee (institutional review board no. 00006761-M2018251). Because it was a retrospective and noninterventive study, informed consent was waived.

Study Sample

Consecutive patients with a confirmed pathologic diagnosis of a primary spinal tumor who underwent surgery at either of two hospitals between March 2010 and October 2022 were identified retrospectively. A total of 203 patients were identified from Peking University Third Hospital, data from whom were temporally split into a training set ($n = 155$; March 2010 to August 2019) and an internal test set ($n = 48$; September 2019 to October 2022) based on the date of diagnosis. An external test set was

collected from the Affiliated Hospital of Qingdao University ($n = 56$; March 2010 to October 2022).

The inclusion criteria were (a) pathologic diagnosis with solitary primary spinal tumor; (b) treatment with surgical tumor resection (gross total resection or subtotal resection); and (c) spinal MRI examination within 4 weeks before treatment. The main exclusion criteria were (a) patients with benign bone tumors who underwent surgery for pain relief; (b) patients with spinal metastases who underwent palliative decompressive surgery; (c) patients who received other therapy (chemotherapy, radiation therapy, interventional treatment) before MRI examination; and (d) postoperative follow-up less than 24 months. The flowchart summarizing patient selection is shown in Figure 1.

Clinical and Tumor Characteristics

Patient baseline clinical characteristics and pathologic data were collected from medical records and histopathologic reports, including age, sex, tumor histologic type and stage, location, International Statistical Classification of Diseases and Related Health Problems grade, volume (whole or paravertebral tumor), Bilsly stage, surgery method, adjuvant therapy (radiation therapy or chemotherapy), postoperative recurrence, and PFS. The assessment criteria are provided in Appendix S1, and the workflow for paraspinal tumor volume measurement is illustrated in Figure S1. The study endpoint was PFS, which was defined as the time from the initiation of treatment to first disease progression (recurrence and metastasis), death from any cause, or last follow-up, whichever came first. Disease progression was confirmed with biopsy and/or imaging, including CT, MRI, or fluorodeoxyglucose PET/CT.

MRI Protocols

All patients underwent preoperative MRI within 4 weeks of surgery. The MRI examination was performed on various systems at the two institutions, including scanners from GE HealthCare,

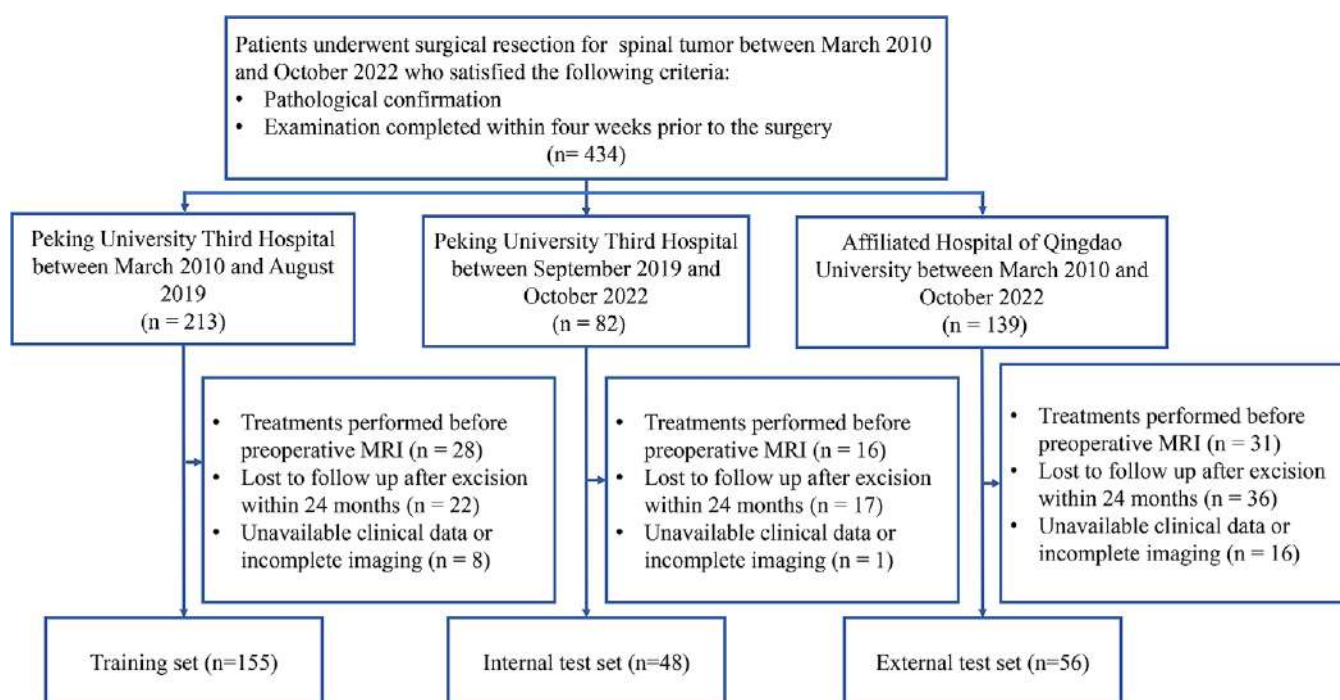


Figure 1: Flowchart summarizes the study sample.

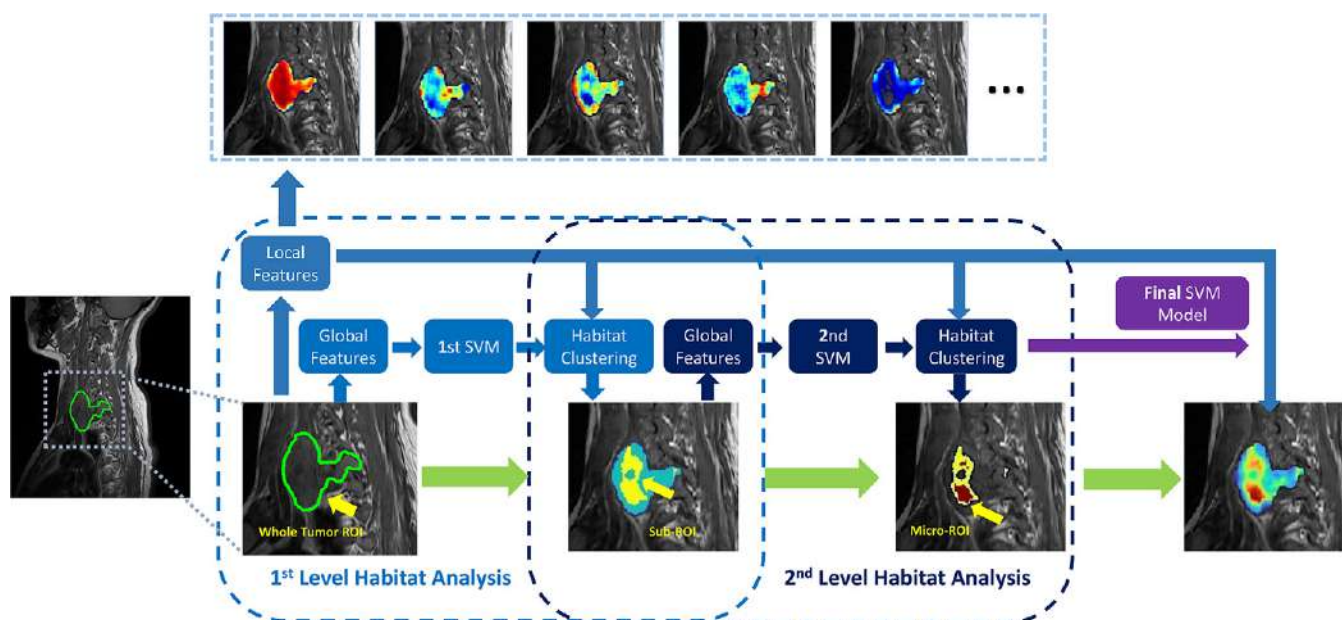


Figure 2: Diagram shows the workflow of the two-iteration nested habitat analysis. The input for the nested habitat analysis is the MRI features within the original region of interest (ROI). The output of the first iteration habitat analysis is the subregion, which serves as the input for the second iteration habitat analysis to extract the microregion. The features from the microregion are then used to build the final support vector machine (SVM) model. Five local feature maps (from left to right: Long Run Low Gray Level Emphasis, Size Zone Non-Uniformity, Cluster Prominence, Small Area High Gray Level Emphasis, and Run Entropy) are shown as examples at the top.

Siemens, Philips, and United Imaging at 1.5 or 3.0 T. The protocol included T1- and T2-weighted sequences, and detailed information about the scanner and imaging parameters is listed in Appendix S1 and Table S1. Because the MRI examination was performed for preoperative assessment, MRI contrast agents were not routinely given, and only noncontrast T1- and T2-weighted images were analyzed.

Traditional Radiomics Analysis

Detailed methods are provided in Appendix S2. The whole-tumor region of interest (ROI) was manually segmented on T1-weighted images by two radiologists (Q.W. and Y.C., with 6 and 12 years of experience in musculoskeletal radiology, respectively). Then, the ROI was mapped to T2-weighted images and modified as necessary. The final ROI was verified by a senior radiologist (N.L., with 19 years of experience in musculoskeletal radiology). A subset was used to evaluate intra- and interrater agreement. Radiomics features were extracted using PyRadiomics (18). Sequential feature selection was used to distill the extracted radiomics features down to the most predictive features. Then, a support vector machine model was developed for the classification of prognosis. The importance of the selected features was evaluated using SHapley Additive exPlanations (hereafter, Shapley) values, which is a common method for interpreting machine learning models by assigning importance values to each feature.

Nested Habitat Analysis

Nested habitat analysis was used to evaluate the heterogeneity within the tumor (Fig 2). A detailed description of the procedures is provided in Appendix S3. Briefly, after the traditional radiomics analysis, further local analysis within the whole-tumor ROI was conducted. Each ROI was divided into smaller patches ($1 \times 1 \times 1$ cm) to extract local features, focusing on texture details and intensity variations. These local features were input into a

support vector machine model to generate probability maps indicating areas of aggressiveness. With use of *k*-means clustering, two subregions were identified as high- versus low-risk tumor areas associated with different prognoses.

Within the high-risk subregion, global features were extracted, and a second support vector machine model was built to assess the probability of aggressiveness. Again, local features were extracted using patches (with the same pixel size of 10×10) to generate corresponding probability maps of high-risk subregions. *k*-means clustering was used to segment each subregion into two microregions based on the probabilities. The high-risk microregion was used to construct the final support vector machine classification model. The same method was applied to the internal and external test sets.

Statistical Analysis

Univariable Cox proportional hazards regression analyses were first performed to assess the association between each clinical and radiomics variable and PFS. Variables with $P < .10$ in the univariable analysis were included in the multivariable Cox regression model. The model's effectiveness was gauged using the hazard ratio (HR) and 95% CI. R software (version 3.6.2, www.r-project.org) was used for statistical analysis, with two-sided $P < .05$ representing statistically significant difference.

Receiver operating characteristic analysis was performed using the area under the receiver operating characteristic curve (AUC) to evaluate the classification performance, and the AUCs of different models (clinical, traditional radiomics, and nested habitat) were compared using the DeLong test. In addition, AUCs were calculated separately for malignant and intermediate-grade cases to evaluate model performance within these pathologic subgroups. Kaplan-Meier survival curves were constructed, and the log-rank test was used to determine the statistical significance of the observed differences in PFS. A post hoc power analysis was

Table 1: Clinical and Pathologic Characteristics of All Patients

Characteristic	Total (n = 259)	Training Set (n = 155)	Internal Test Set (n = 48)	External Test Set (n = 56)	P Value
Sex					.75
F	150	92	28	30	
M	109	63	20	26	
Age (y)*	39 (28–53)	38 (26–53)	38 (30–52)	42 (32–55)	.89
Location					<.001 [†]
Cervical	103	71	25	7	
Thoracic	77	47	14	16	
Lumbar	62	32	9	21	
Sacrococcygeal	17	5	0	12	
Enneking stage					.02 [†]
II	108	71	23	14	
III	151	84	25	42	
Surgical method					<.001 [†]
Total vertebral resection	198	128	40	30	
Intralesional resection	61	27	8	26	
Extent of resection					.003 [†]
Gross total resection	215	137	41	37	
Subtotal resection	44	18	7	19	
Vertebral compression					<.001 [†]
Yes	107	66	26	15	
No	152	89	22	41	
Multivertebral involvement					.31
No	136	78	30	28	
Yes	123	77	18	28	
ICD grade					.43
Intermediate	117	73	23	21	
Malignant	142	82	25	35	
Bilsky grade					.21
0	120	76	21	23	
1	89	49	20	20	
2	39	20	7	12	
3	11	10	0	1	
Histologic classification					<.001 [†]
Chondrogenic	30	14	10	6	
Osteogenic	30	11	10	9	
Fibrogenic	2	2	0	0	
Vascular tumors of bone	7	7	0	0	
Osteoclastic giant cell-rich	91	63	15	13	
Notochordal	54	24	13	17	
Other mesenchymal	12	12	0	0	
Hematopoietic	19	9	0	10	
Miscellaneous	14	13	0	1	
Three-year progression					.46
No	183	114	32	37	
Yes	76	41	16	19	
Follow-up time (mo)*	26 (16–40)	24 (14–48)	24 (17–35)	29 (24–42)	<.001 [†]

Note.—ICD = International Statistical Classification of Diseases and Related Health Problems.

* Data are medians, with IQRs in parentheses.

[†] Represents statistically significant difference ($P < .05$). P values represent comparisons among the training, internal test, and external test sets, calculated using the χ^2 test for categorical variables and the Mann-Whitney U test for continuous variables.

Table 2: Clinical and Pathologic Characteristics of Patients with Good versus Poor Prognosis Determined by 3-year Progression-Free Survival

Characteristic	Training and Internal Test Sets			External Test Set		
	Nonprogression (n = 146)	Progression (n = 57)	P Value	Nonprogression (n = 37)	Progression (n = 19)	P Value
Sex						
F	86	34	.92	18	12	.30
M	60	23		19	7	
Age (y)*	36 (27–51)	41 (28–56)	.25	41 (29–48)	51 (39–66)	.43
Location			<.001 [†]			.32
Cervical	65	31		3	4	
Thoracic	53	8		13	3	
Lumbar	28	13		13	8	
Sacrococcygeal	0	5		8	4	
Enneking stage			.001 [†]			.002 [†]
II	81	13		14	0	
III	65	44		23	19	
Surgical method			.12			.65
Total vertebral resection	120	48		19	11	
Intralesional resection	26	9		18	8	
Extent of resection			.99			.79
Gross total resection	128	50		24	13	
Subtotal resection	18	7		13	6	
Vertebral compression			.004 [†]			.96
Yes	57	35		10	5	
No	89	22		27	14	
Multivertebral involvement			.048 [†]			.16
No	84	24		16	12	
Yes	62	33		21	7	
ICD grade			.06			.07
Intermediate	75	21		17	4	
Malignant	71	36		20	15	
Bilsky grade			.03 [†]			.61
0	79	18		14	9	
1	44	25		15	5	
2	18	9		7	5	
3	5	5		1	0	
Histologic classification			.60			.36
Chondrogenic	17	7		3	3	
Osteogenic	16	5		8	1	
Fibrogenic	2	0		0	0	
Vascular tumors of bone	6	1		0	0	
Osteoclastic giant cell-rich	59	19		10	3	
Notochordal	22	15		10	7	
Other mesenchymal	7	5		0	0	
Hematopoietic	7	2		5	5	
Miscellaneous	10	3		1	0	
Follow-up time (mo)*	30 (23–48)	12 (8–24)	<.001 [†]	36 (29–52)	19 (14–24)	<.001 [†]

Note.—Significant differences between the progression and nonprogression groups were observed in tumor location, Enneking stage, vertebral compression, multivertebral involvement, Bilsky grade, and follow-up time in the training and internal test sets ($P < .05$). In the external test set, significant differences were found in Enneking stage and follow-up time ($P = .002$ and $P < .001$, respectively). ICD = International Statistical Classification of Diseases and Related Health Problems.

* Data are medians, with IQRs in parentheses.

[†] Indicates statistically significant difference ($P < .05$). P values were calculated using the χ^2 test for categorical variables and the Mann-Whitney U test for continuous variables, representing comparisons between the progression and nonprogression groups within each institution.

Table 3: Prediction of Progression-Free Survival Based on Clinical Factors of the Entire Study Sample

Potential Factor	Univariable Analysis		Multivariable Analysis	
	HR	P Value	HR	P Value
Sex	0.77 (0.36, 1.56)	.92	0.72 (0.34, 1.51)	.38
Age	0.95 (0.93, 0.99)	.26	0.95 (0.91, 0.98)	.02*
Location	3.21 (2.51, 4.90)	.001	1.21 (0.62, 1.80)	.06
Volume	0.71 (0.36, 1.06)	.42	0.92 (0.60, 1.24)	.33
Enneking stage	2.61 (2.30, 3.21)	.001	1.85 (1.30, 4.52)	.04*
Surgical method	1.94 (1.01, 2.90)	.11	1.45 (0.60, 3.52)	.41
Vertebral compression	2.71 (2.41, 3.18)	.004	0.27 (0.12, 0.62)	.002*
Multivertebral involvement	1.98 (1.27, 3.34)	.05	2.61 (0.78, 8.80)	.12
ICD-10	2.21 (1.87, 2.68)	.06	1.66 (1.04, 2.64)	.03*
Histologic classification	0.49 (0.34, 0.67)	.60	1.08 (0.89, 1.31)	.44
Bilsky grade	2.03 (1.10, 2.98)	.18	1.82 (0.89, 2.77)	.07
Paravertebral tumor volume	1.17 (1.12, 3.01)	.24	0.94 (0.87, 1.01)	.29
Chemotherapy	1.77 (0.91, 2.93)	.16	1.12 (0.59, 2.07)	.26
Radiation therapy	1.43 (0.62, 3.24)	.35	0.99 (0.49, 1.98)	.40

Note.—Data in parentheses are 95% CIs. In the multivariable Cox regression analysis, older age (hazard ratio [HR], 0.95; $P = .02$); higher Enneking stage (HR, 1.85; $P = .04$); vertebral compression (HR, 0.27; $P = .002$); and International Statistical Classification of Diseases and Related Health Problems, 10th revision (ICD-10), malignancy (HR, 1.66; $P = .03$) were identified as independent predictors of shorter progression-free survival.

* $P < .05$. P values were calculated based on Cox regression models and indicate the strength of association between each factor and progression-free survival.

performed to evaluate the statistical power of the study (Appendix S4). All statistical analyses were performed by two authors (Q.W. and Y.Z.).

Data Availability

The datasets and code used during the current study are available from the corresponding author upon reasonable request.

Results

Patient Characteristics

A total of 259 patients were included (median age, 39 years [IQR, 28–53 years]; 150 female) (Table 1). As shown in Figure 1, among the 434 patients who underwent surgical resection and met the initial inclusion criteria, 175 were excluded due to preoperative treatment before MRI ($n = 75$), follow-up duration less than 24 months ($n = 75$), or incomplete clinical and/or imaging data ($n = 25$). The median PFS time was 26 months (range, 3–100 months) among all patients. Postoperative progression within 3 years occurred in 76 patients, accounting for 29% (76 of 259) (Table 2). Patients in the training and internal test sets had more cervical tumors (47% [96 of 203]), and patients in the external test set had more lumbar and sacrococcygeal tumors (59% [33 of 56]). A total of 83% (168 of 203 patients) in the training and internal test sets received total vertebral resection, compared with 54% (30 of 56 patients) in the external test set. Subtotal resection was performed more frequently in the external test set (34% [19 of 56])

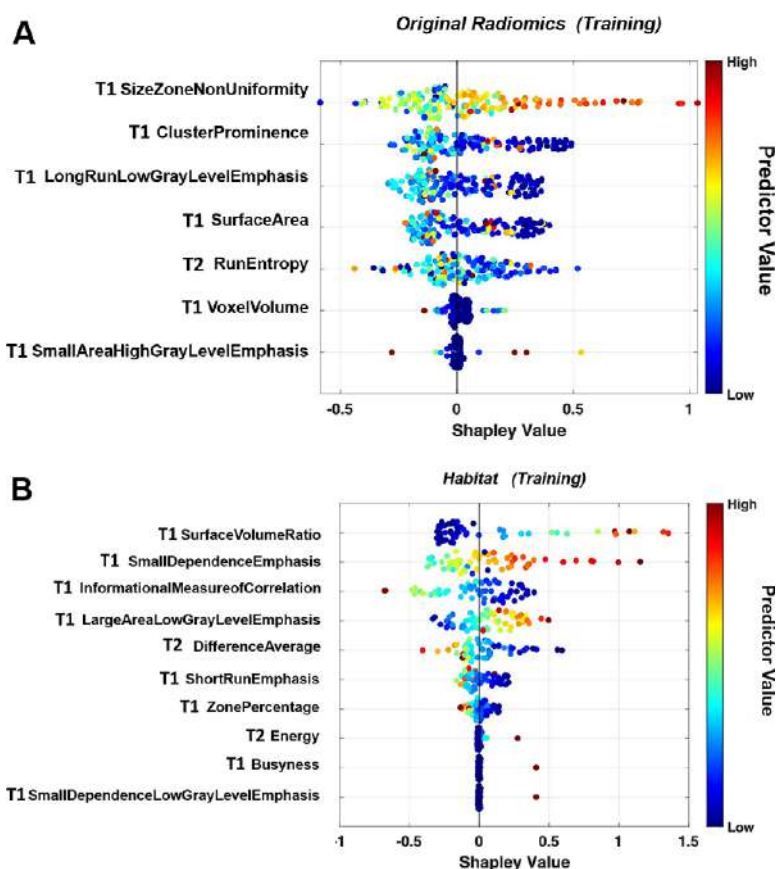


Figure 3: Feature importance for progression prediction derived from traditional radiomics and habitat models. SHapley Additive exPlanations (ie, Shapley) plots show selected features from (A) traditional radiomics analysis ($n = 7$) and (B) nested habitat analysis ($n = 10$) for interpreting the importance of features for predicting progression. The x-axis represents the Shapley value, which indicates the impact of each feature on the model's prediction. The y-axis lists the selected features from the respective models. Each point represents an individual patient. The color bar reflects the original value of the feature: Blue indicates low values, and red indicates high values. Most selected features are derived from T1-weighted MRI scans.

than in the training and internal test sets (12% [25 of 203]). The difference between patients who experienced progression versus nonprogression across the two institutions is listed in Tables 2 and S5. Apparent differences between the progression and nonprogression groups were observed in the training and internal test sets. Patients in the progression group had a higher proportion of Enneking stage III tumors (77% [44 of 57] vs 45% [65 of 146]; $P = .001$), greater vertebral compression (61% [35 of 57] vs 39% [57 of 146]; $P = .004$), and higher Bilsky grade (ie, 2–3) (25% [14 of 57] vs 16% [23 of 146]; $P = .03$). Further comparisons of tumor volume and adjuvant therapy are provided in Appendix S5 (Tables S4, S5).

Risk Factors for Progression and Development of the Clinical Model

The corresponding HRs of each clinical factor from a PFS predictive clinical factor analysis of all patients are shown in Table 3. Multivariable analysis showed that patients with Enneking stage III (HR, 1.85 [95% CI: 1.30, 4.52]; $P = .04$); vertebral compression (HR, 0.27 [95% CI: 0.12, 0.62]; $P = .002$); and high-grade International Statistical Classification of Diseases and Related Health Problems, 10th revision, classification (HR, 1.66 [95% CI: 1.04, 2.64]; $P = .03$) were more likely to show tumor progression. The clinical model subsequently included these three factors.

Development of Radiomics Models

A total of 1299 radiomics features were extracted from T1- and T2-weighted images of each patient, of which 1030 were stable features. Following the Shapley analysis, the top seven radiomics features (six from T1 and one from T2) were selected to build the traditional model,

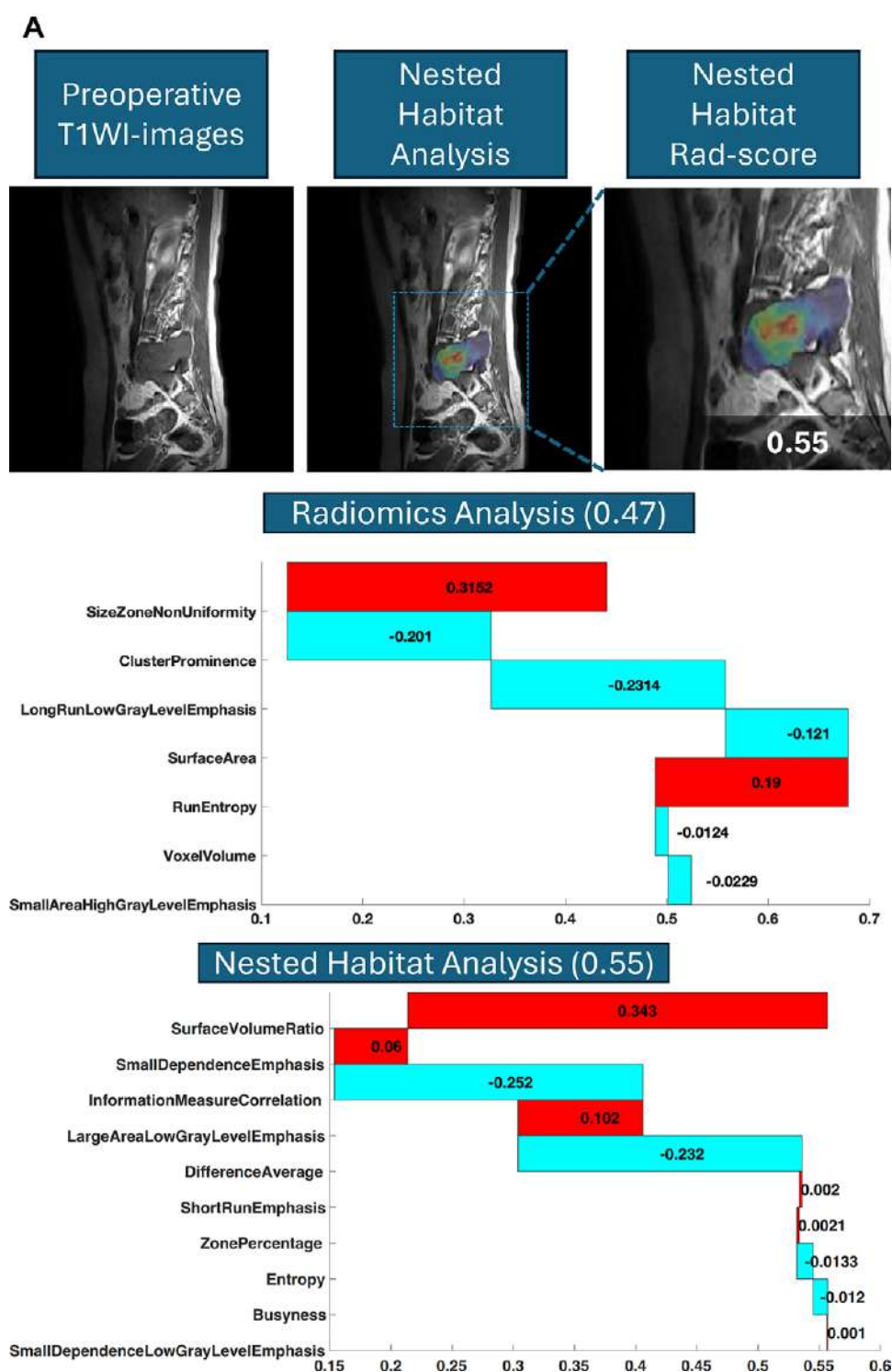


Figure 4: Examples of progression prediction using radiomics and nested habitat models based on preoperative noncontrast sagittal T1-weighted (T1WI) MRI scans, shown using waterfall plots. On the far left at the top of each panel are representative T1-weighted images from each of the three patients. In the middle image of each set, a blue box outlines the segmented tumor, and the overlaid color map indicates spatial heterogeneity within the lesion. The top waterfall plot depicts the SHapley Additive exPlanations (ie, Shapley) values for the seven selected radiomics features. The x-axis shows Shapley values indicating the impact on prediction; the y-axis lists individual features. Red bars indicate positive contributions to progression prediction, and blue bars indicate negative contributions. The bottom plot shows Shapley values for the 10 selected habitat features from the nested analysis, with a similar visualization. **(A)** A 41-year-old female patient with a lumbar giant cell tumor and postoperative recurrence. Predicted probability of progression: 0.47 by the radiomics model (false negative) and 0.55 by the habitat model (true positive) (Fig 4 continues).

and the top 10 microregion features (eight from T1 and two from T2) were selected to build the nested habitat model. The selected features and their Shapley values are shown in Figure 3. The radiomics scores obtained from three different patient examples are shown in Figure 4. Microregions were identified in a small region within the tumor, approximately 30% of the whole tumor. The volume range of microregions was from 15 to 50 146 mm³, representing 4%–52% of the whole-tumor volume.

Comparison of Model Performance for PFS Estimation

The performance for predicting 3-year PFS using the clinical and two radiomics models and the two combined radiomics-clinical models across each dataset is shown in Table 4. In the training set, both the traditional radiomics score (HR, 2.80; $P = .03$) and habitat radiomics score (HR, 3.50; $P = .01$) were independent predictors of shorter PFS after surgery. When the clinical factors (Enneking stage III; vertebral compression; and high-grade International Statistical Classification of Diseases and Related Health Problems, 10th revision, classification) were included, the AUC of the tumor radiomics model increased from 0.82 (95% CI: 0.68, 0.93) to 0.86 (95% CI: 0.78, 0.94) in the training set. The AUC of the habitat radiomics model increased from 0.93 (95% CI: 0.87, 0.98) to 0.95 (95% CI: 0.88, 0.99). Although numerically higher, this difference was not statistically significant ($P = .93$) (Table 4).

In the comparison of receiver operating characteristic curves using the DeLong test in the training set, the combined habitat and clinical model outperformed the clinical model (AUC of combined habitat and clinical model, 0.95 [95% CI: 0.88, 0.99] vs AUC of clinical model, 0.75 [95% CI: 0.65, 0.85]; $P = .004$) and the traditional combined tumor and clinical model (AUC, 0.86 [95% CI: 0.78, 0.94]; $P = .04$). However, in the internal test set, although the habitat and clinical model achieved a higher AUC (0.86 [95% CI: 0.69, 0.96]) than

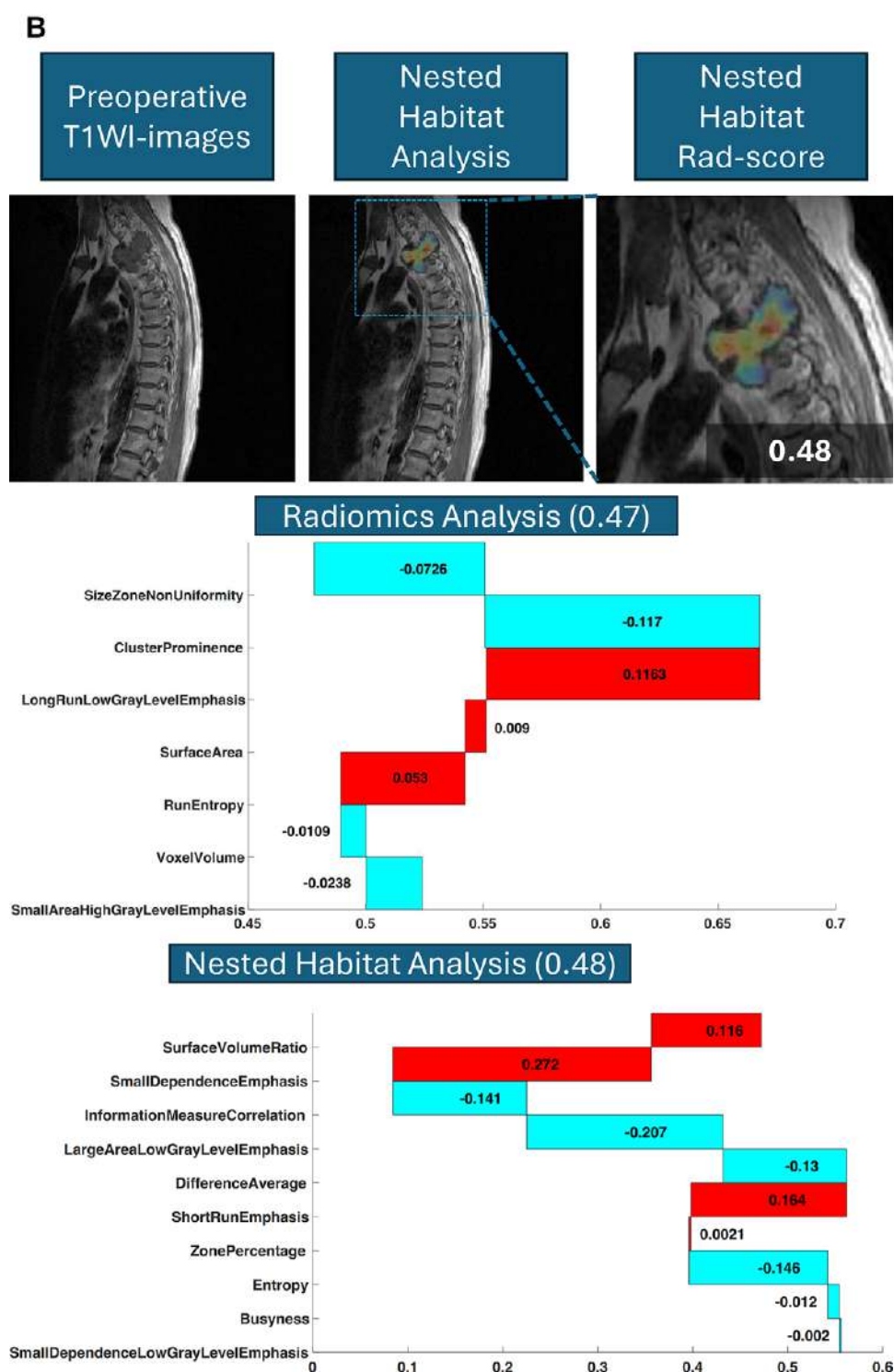


Figure 4 (continued): (B) A 52-year-old male patient with thoracic chordoma without recurrence. Predicted probability: 0.47 (radiomics) and 0.48 (habitat), both true-negative predictions (Fig 4 continues).

the clinical model (AUC, 0.66 [95% CI: 0.52, 0.80]; $P = .04$), no statistical significance was observed in comparisons with the tumor radiomics (AUC, 0.80 [95% CI: 0.65, 0.93]; $P = .62$) or traditional combined models (AUC, 0.82 [95% CI: 0.67, 0.94]; $P = .67$), presumably due to a smaller number of cases ($n = 48$) and shorter follow-up times. In the external test set, the clinical model (AUC, 0.70 [95% CI: 0.55, 0.85]) had a lower performance compared with the habitat and clinical model ($P = .04$).

The traditional radiomics (AUC, 0.81 [95% CI: 0.68, 0.94]; $P = .34$) and the combined tumor and clinical models (AUC, 0.81 [95% CI: 0.68, 0.94]; $P = .34$) showed no significant differences relative to the habitat and clinical model. Additionally, receiver operating characteristic analysis was performed separately for malignant and intermediate cases (Appendix S6; Tables S6, S7), and neither the traditional radiomics nor the combined tumor and clinical model significantly predicted PFS in the external test set (malignant group, $P = .44$ for both models [AUC, 0.83 {95% CI: 0.70, 0.95}] and intermediate group, $P = .45$ and .49 [AUCs, 0.79 {95% CI: 0.66, 0.92} and 0.80 {95% CI: 0.69, 0.95}], respectively).

Differences in the progression probabilities predicted by the traditional radiomics versus the habitat model are shown in waterfall plots for each dataset (Fig 5A–5C), stratified according to the 3-year PFS status. In the training set, the mean difference was 0.14 for patients with progression compared with -0.03 for those without progression ($P < .001$). In the internal validation set, the mean difference was 0.13 for patients with progression compared with 0 for nonprogression ($P = .003$). Similarly, in the external validation set, the mean difference was 0.14 for patients with progression and -0.04 for nonprogression ($P < .001$). A greater difference was observed in patients showing progression for each of the training and test sets, suggesting that the habitat analysis could increase the predicted progression probability. The receiver operating characteristic curves of the clinical, habitat radiomics, and combined models are shown in Figure 5D–5F.

As shown in Figure 6A–6C, the Kaplan-Meier curves based on the clinical, traditional radiomics, and habitat models in the training set showed significant differences in PFS between high- and low-risk groups, with a median PFS of 34 versus 24 months (log-rank $P = .01$), 30 versus 24 months ($P = .03$), and 31 versus 22 months ($P = .01$), respectively. In the external test set (Fig 6D–6F), only the performance of the habitat model remained statistically significant, with a median PFS of 28

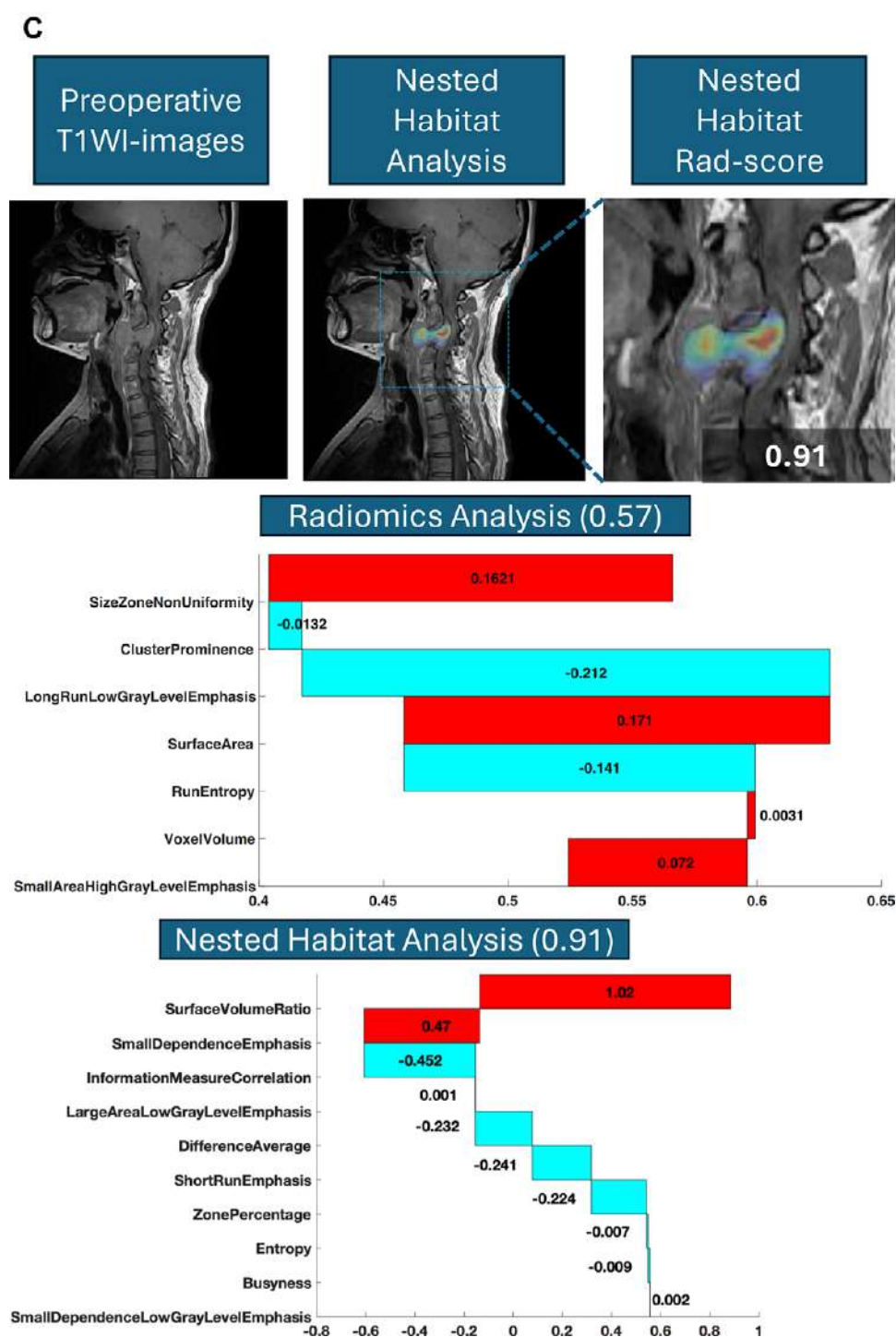


Figure 4 (continued): (C) A 38-year-old female patient with a cervical giant cell tumor and recurrence. Predicted probability: 0.57 (radiomics) and 0.91 (habitat), both true-positive predictions, with the habitat model giving a higher predicted probability.

versus 24 months (log-rank $P = .04$), whereas the performance of the clinical and radiomics models showed no evidence of a difference (median PFS, 26 vs 24 months [$P = .16$] and 25 vs 24 months [$P = .36$], respectively). In the external test set, only the habitat model significantly stratified PFS in both malignant and intermediate tumors (median PFS, 31 vs 25 and 26 vs 21 months, respectively; HR, 3.4 and 3.04; $P = .02$ and .04) (Figs S2, S3), whereas clinical and traditional radiomics models showed no significant separation ($P > .05$).

Table 4: Prediction Performance of Progression-Free Survival According to Different Clinical and Radiomics Models

Model	Training Set			Internal Test Set			External Test Set		
	AUC	z Value	P Value*	AUC	z Value	P Value*	AUC	z Value	P Value*
Clinical	0.75 (0.65, 0.85)	3.79	.004	0.66 (0.52, 0.80)	1.74	.04	0.70 (0.55, 0.85)	2.01	.04
Radiomics tumor	0.82 (0.68, 0.93)	2.80	.03	0.80 (0.65, 0.93)	0.59	.62	0.81 (0.68, 0.94)	0.95	.34
Combined tumor and clinical	0.86 (0.78, 0.94)	1.97	.04	0.82 (0.67, 0.94)	0.42	.67	0.81 (0.68, 0.94)	0.95	.34
Radiomics habitat	0.93 (0.87, 0.98)	0.08	.93	0.86 (0.69, 0.96)	0.01	.99	0.88 (0.75, 0.96)	0.02	.97
Combined habitat and clinical	0.95 (0.88, 0.99)	NA	NA	0.86 (0.69, 0.96)	NA	NA	0.89 (0.76, 0.96)	NA	NA

Note.—Data in parentheses are 95% CIs. AUC = area under the receiver operating characteristic curve, NA = not applicable.

* P values were calculated using the DeLong test, comparing each model with the combined habitat and clinical model.

Discussion

Accurate preoperative prognostic assessment of aggressive spinal tumors is essential for optimal treatment planning. In this study, we applied nested habitat radiomics analysis to develop MRI models for predicting 3-year progression-free survival (PFS) in patients with primary spinal tumors. Habitat analysis generated probability maps associated with prognosis within a heterogeneous tumor reflecting multiregional imaging phenotypes. The combined habitat and clinical model achieved areas under the receiver operating characteristic curves (AUCs) of 0.95 in the training set, 0.86 in the internal test set, and 0.89 in the external test set. In Kaplan-Meier analysis, the habitat-based model stratified patients by progression risk in the external test cohort (median PFS, 28 vs 24 months; log-rank $P = .04$), whereas the clinical (median PFS, 26 vs 24 months; $P = .16$) and traditional radiomics (median PFS, 25 vs 24 months; $P = .36$) models did not show evidence of a statistically significant difference.

Spinal tumors comprise a heterogeneous group of pathologic entities, many of which cannot be readily classified as strictly benign or malignant. Although some tumor types, such as osteoblastoma, aneurysmal bone cyst, and giant cell tumor, were considered benign for many years, they are now considered invasive with metastatic potential. Thus, for both malignant and intermediate “aggressive” tumors, surgical excision is the dominant treatment option. Conventionally, the histologic types of the tumor should be determined first, followed by an analysis of the prognostic factors. However, this strategy faces significant challenges for spinal tumors. Identifying tumor types based on radiologic or even pathologic examinations may introduce errors due to the highly heterogeneous nature of spinal lesions and the difficulty of biopsies in targeting the most aggressive area. MRI provides a good assessment of the tumor tissue composition, morphologic characteristics, and bone and surrounding soft tissue involvement. Most locally aggressive and malignant vertebral tumors exhibit soft tissue masses, subtle infiltration, mixed T1- and T2-weighted imaging signals, necrosis, osteogenesis, and cartilage matrix. Many overlapping features are noted across different levels of invasion, and understanding their subtleties may increase the chance of success in achieving complete excision.

Radiomics analysis has been applied to differentiate various spinal tumors (19–23), such as chondrosarcoma versus chordoma. The nested habitat analysis method offers a new

approach for subregional analysis by means of sophisticated radiomics features and advanced machine learning techniques. Our results show that the hierarchical segmentation offered by the habitat analysis can localize parts of the tumor that are related to poor prognosis. In particular, the improvement in AUC over the clinical model in the external test set was statistically significant, whereas differences compared with other models were not. This suggests potential robustness and translational value of the proposed nested habitat analysis method, although further validation is warranted. In the external test set, the combined habitat and clinical model achieved the highest AUC of 0.89 but did not significantly outperform the radiomics model ($P = .34$). However, both models showed better performance than the clinical model ($P = .04$). This suggests that the nested habitat model can identify imaging patterns that are consistent across institutions. The performance in the external test set supports its potential value in detecting high-risk microregions within the tumor using a probability-guided approach. Traditional models based on whole-tumor features may overlook the contribution of aggressive subregions, whereas the nested habitat model may allow for a more biologically informed risk assessment.

Several studies have applied habitat analysis strategies that involve delineating and clustering tumor subregions to gain insights into tumor behavior and unravel the heterogeneity of malignant tumors (24). For example, preoperative MRI habitat analysis can predict glioblastoma survival (25) and tumor progression after radiation therapy (26,27) and distinguish gene phenotypes (28). Similar research has been performed for other malignant neoplasms, including breast cancer (29) and hepatocellular carcinoma (30–32). These studies indicate that habitat analysis facilitates the localization of different tumor patches within a heterogeneous tumor to understand how they contribute to assessing pathologic risk factors and predicting treatment response and prognosis. Most of these studies directly used the anatomic structures (eg, tumor and peritumoral regions), multiparametric MRI features (eg, enhancing tissue, edema at fluid-attenuated inversion-recovery imaging, necrosis, apparent diffusion coefficient, cellular density), or the radiomics features to separate habitats. Our study uniquely used the risk probability maps from radiomics models to identify habitats and, through two iterations, identified high-risk microregions, accounting for approximately 30% of the tumor volume.

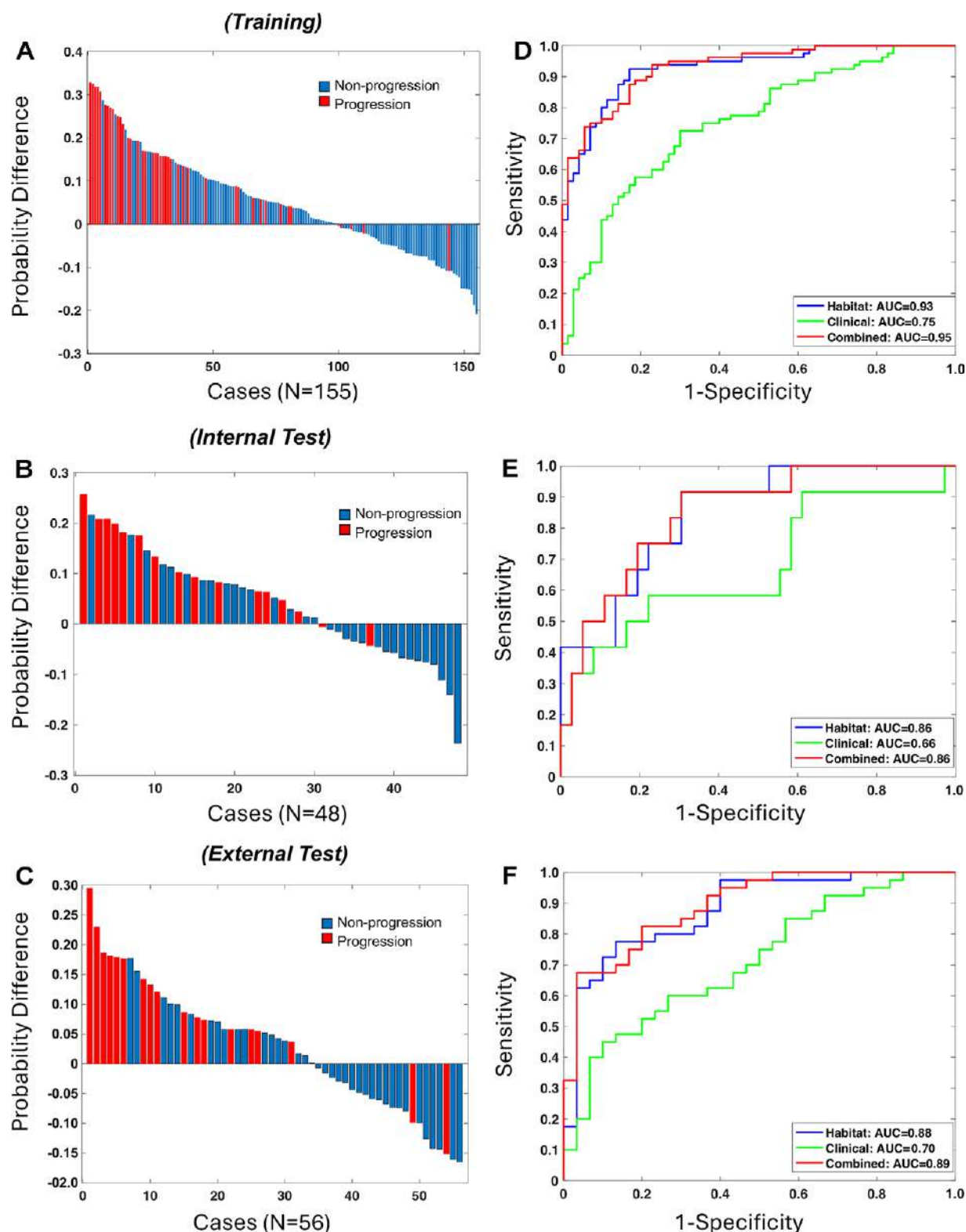


Figure 5: Difference in predicted progression probabilities and model performance across datasets. **(A–C)** Waterfall plots show the difference in predicted progression probabilities between the habitat and traditional radiomics models in the **(A)** training, **(B)** internal test, and **(C)** external test sets. Each bar represents one patient. The x-axis indicates individual cases; the y-axis shows the difference in predicted progression probability (habitat model minus radiomics model). Red bars represent patients who experienced progression at 3 years; blue bars represent nonprogression patients. A greater positive difference suggests improved predictive value of the habitat model in patients who experienced progression. **(D–F)** Receiver operating characteristic curves for clinical (green), habitat radiomics (blue), and combined habitat and clinical (red) models in the **(D)** training, **(E)** internal test, and **(F)** external test sets. The x-axis shows 1 minus specificity; the y-axis shows sensitivity. AUC = area under the receiver operating characteristic curve.

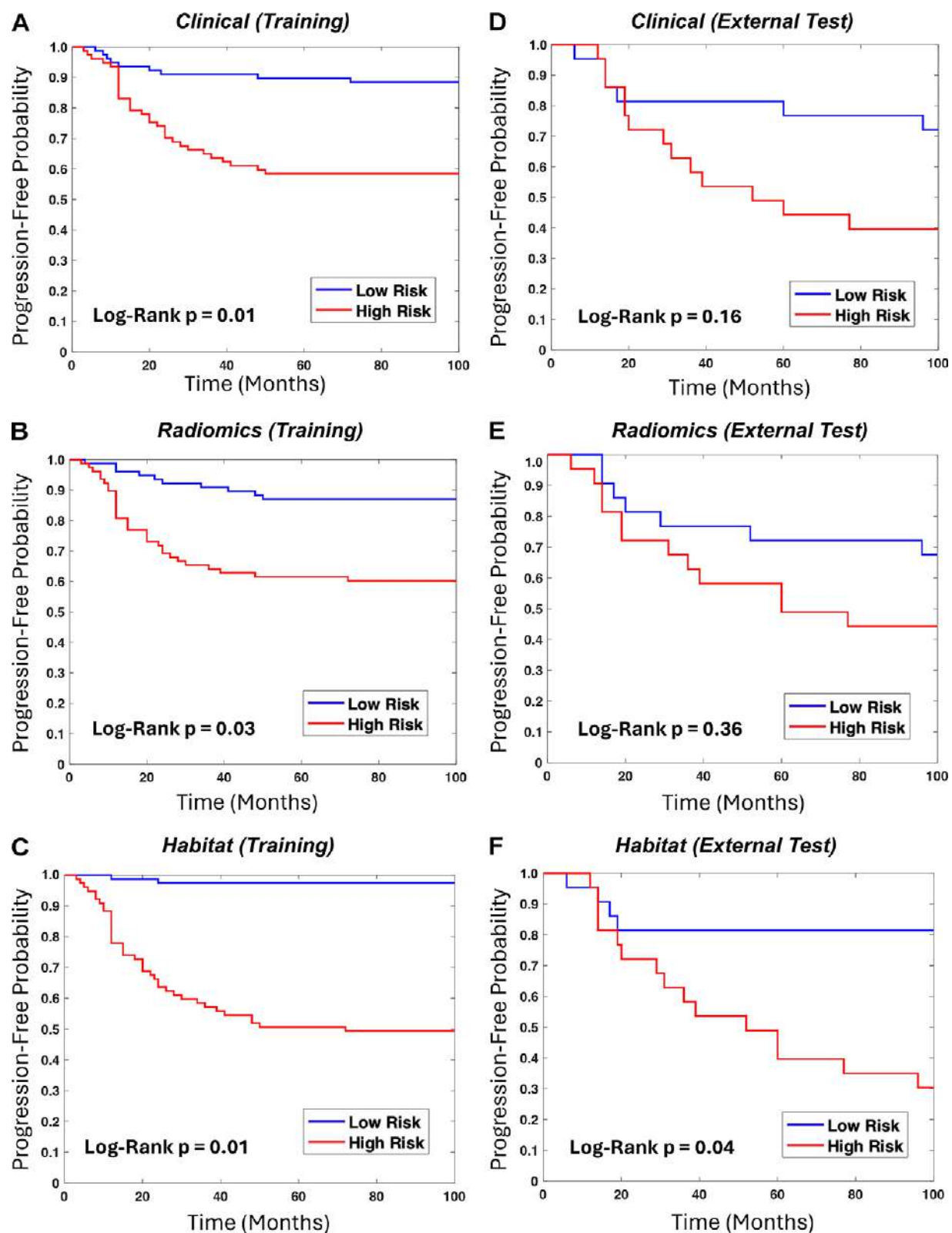


Figure 6: Kaplan-Meier curves for progression-free survival using the (A, D) clinical, (B, E) traditional radiomics, and (C, F) nested habitat models. A–C show results in the training set, and D–F in the external test set. In each subplot, the x-axis indicates follow-up time in months, and the y-axis represents progression-free probability. Patients are stratified into high-risk (red curve) and low-risk (blue curve) groups. The *P* value in each panel was calculated using the log-rank test. The habitat model shows the most significant separation in both training ($P = .01$) and external test ($P = .04$) sets.

Although the treatment of spinal tumors has changed substantially over the past few decades, the multidisciplinary decision-making process is hampered by the lack of individualized prognostic assessment (9). Surgical resection decision-making needs to weigh tumor malignancy, recurrence probability, spinal stability, nerve function, and the risk of postoperative complications to determine the scope of resection (33). In this dilemma, surgeons need to evaluate tumor aggressiveness to determine whether the benefit is worth the risk. In patients with a high probability of recurrence, postsurgical radiation therapy and frequent radiologic follow-up can be performed (34). Unfortunately, there are no established markers for early risk stratification. The histopathologic diagnosis based on a limited biopsy sample may not provide reliable information. If the most aggressive tumors can be identified with habitat analysis at preoperative MRI and the habitat analysis results can also guide biopsy, this can inform the selection of optimal surgery and postsurgical treatment to reduce the risk of progression.

Other artificial intelligence-based image analysis methods, such as deep learning using vision transformer (35) or Grad-CAM (36), are also capable of identifying attention regions on images. However, deep learning requires very large datasets and computational resources. The habitat method is data-efficient, and it can be widely implemented in real-world clinical environments.

Our study has several limitations. Because this study only included patients with solitary spinal tumors who underwent surgical treatment, the sample size remained relatively small despite being drawn from two tertiary spine centers, which could potentially limit the representativeness of the cohort. Second, there were differences in clinical characteristics, surgical methods, and PFS across the two hospitals; however, this is common in real-world settings. Although selection bias was inevitable, the trained models achieved good results in the external test set. Third, some molecular biomarkers have been used to predict the prognosis of spinal tumors. However, there is no consensus on which tumor needs to be tested, and there is no routine clinical testing procedure. Fourth, four patients in the cohort died during postoperative follow-up, but in the presence of coexisting diseases, it was not possible to determine whether this was due to the tumor, which may lead to data bias. Finally, the ROI was contoured manually due to the complexity of adjacent anatomy, which could be improved with artificial intelligence tools in the future.

In conclusion, in this retrospective study of patients with primary spinal tumors, a nested habitat radiomics model outperformed other models in predicting postoperative progression-free survival from preoperative MRI. This method may help target biopsy and select optimal treatment plans for patients with aggressive spinal tumors.

Deputy Editor: John Carrino

Scientific Editor: Kate Vilas

Author affiliations:

¹ Department of Radiology, Peking University Third Hospital, 49 North Garden Rd, Haidian District, 100069 Beijing, PR China

² Department of Radiation Oncology, University of California–Irvine, Irvine, Calif

³ Department of Radiology, the Affiliated Hospital of Qingdao University, Qingdao, PR China

⁴ Department of Radiological Sciences, Tu & Yuen Center for Functional Onco-Imaging, University of California–Irvine, Irvine, Calif

⁵ Department of Medical Imaging and Radiological Sciences, Kaohsiung Medical University, Kaohsiung, Taiwan

⁶ State Key Laboratory of Vascular Homeostasis and Remodeling, Peking University Third Hospital, Beijing, PR China

Received October 4, 2024; revision requested November 18; final revision received August 26, 2025; accepted September 19.

Address correspondence to: N.L. (email: langning800129@126.com).

Funding: National Natural Science Foundation of China (grant no. 82371921).

Author contributions: Guarantors of integrity of entire study, Q.W., Y.Z., Y.C., W.Z., D.H., M.Y.S., N.L.; study concepts/study design or data acquisition or data analysis/interpretation, all authors; manuscript drafting or manuscript revision for important intellectual content, all authors; approval of final version of submitted manuscript, all authors; agrees to ensure any questions related to the work are appropriately resolved, all authors; literature research, Q.W., Y.Z., T.W., R.Y., K.L., W.Z., N.L.; clinical studies, Q.W., Y.Z., Y.C., R.Y., K.L., W.Z., D.H., M.Y.S., N.L.; experimental studies, Y.Z., R.Y., K.L., W.Z., M.Y.S.; statistical analysis, Q.W., Y.Z., R.Y., K.L., W.Z., M.Y.S.; and manuscript editing, Q.W., Y.Z., R.Y., K.L., W.Z., M.Y.S., N.L.

Disclosures of conflicts of interest: Q.W. No relevant relationships. Y.Z. No relevant relationships. T.W. No relevant relationships. Y.C. No relevant relationships. R.Y. No relevant relationships. K.L. No relevant relationships. W.Z. No relevant relationships. D.H. No relevant relationships. M.Y.S. No relevant relationships. N.L. No relevant relationships.

References

1. Sundaresan N, Rosen G, Boriani S. Primary malignant tumors of the spine. *Orthop Clin North Am* 2009;40(1):21–36, v.
2. Missenard G, Bouthors C, Fadel E, Court C. Surgical strategies for primary malignant tumors of the thoracic and lumbar spine. *Orthop Traumatol Surg Res* 2020;106(1S):S53–S62.
3. Melcher I, Disch AC, Khodadadyan-Klostermann C, et al. Primary malignant bone tumors and solitary metastases of the thoracolumbar spine: results by management with total en bloc spondylectomy. *Eur Spine J* 2007;16(8):1193–1202.
4. Sciubba DM, De la Garza Ramos R, Goodwin CR, et al. Total en bloc spondylectomy for locally aggressive and primary malignant tumors of the lumbar spine. *Eur Spine J* 2016;25(12):4080–4087.
5. Gregory TM, Coriat R, Mir O. Prognostic scoring systems for spinal metastases in the era of anti-VEGF therapies. *Spine* 2013;38(11):965–966.
6. Tokuhashi Y, Matsuzaki H, Toriyama S, Kawano H, Ohsaka S. Scoring system for the preoperative evaluation of metastatic spine tumor prognosis. *Spine* 1990;15(11):1110–1113.
7. Fournier DR, Frangou EM, Ryken TC, et al. Spinal instability neoplastic score: an analysis of reliability and validity from the spine oncology study group. *J Clin Oncol* 2011;29(22):3072–3077.
8. Tomita K, Kawahara N, Kobayashi T, Yoshida A, Murakami H, Akamaru T. Surgical strategy for spinal metastases. *Spine* 2001;26(3):298–306.
9. Szövérfi Z, Lazary A, Bozsódi Á, Klemencsics I, Éltes PE, Varga PP. Primary Spinal Tumor Mortality Score (PSTMS): a novel scoring system for predicting poor survival. *Spine J* 2014;14(11):2691–2700.
10. Tsukamoto S, Mavrogenis AF, van Langevelde K, van Vucht N, Kido A, Errani C. Imaging of spinal bone tumors: principles and practice. *Curr Med Imaging* 2022;18(2):142–161.
11. Bera K, Braman N, Gupta A, Velcheti V, Madabhushi A. Predicting cancer outcomes with radiomics and artificial intelligence in radiology. *Nat Rev Clin Oncol* 2022;19(2):132–146.
12. Sun R, Limkin EJ, Vakalopoulou M, et al. A radiomics approach to assess tumour-infiltrating CD8 cells and response to anti-PD-1 or anti-PD-L1 immunotherapy: an imaging biomarker, retrospective multicohort study. *Lancet Oncol* 2018;19(9):1180–1191.
13. Pinker K, Chin J, Melsaether AN, Morris EA, Moy L. Precision medicine and radiogenomics in breast cancer: new approaches toward diagnosis and treatment. *Radiology* 2018;287(3):732–747.
14. Gillies RJ, Kinahan PE, Hricak H. Radiomics: images are more than pictures, they are data. *Radiology* 2016;278(2):563–577.
15. Waqar M, Van Houdt PJ, Hessen E, et al. Visualising spatial heterogeneity in glioblastoma using imaging habitats. *Front Oncol* 2022;12:1037896.
16. Beig N, Bera K, Prasanna P, et al. Radiogenomic-based survival risk stratification of tumor habitat on Gd-T1w MRI is associated with biological processes in glioblastoma. *Clin Cancer Res* 2020;26(8):1866–1876.

17. Tabassum M, Suman AA, Suero Molina E, Pan E, Di Ieva A, Liu S. Radiomics and machine learning in brain tumors and their habitat: a systematic review. *Cancers (Basel)* 2023;15(15):3845.
18. van Griethuysen JJM, Fedorov A, Parmar C, et al. Computational radiomics system to decode the radiographic phenotype. *Cancer Res* 2017;77(21):e104-e107.
19. Yamazawa E, Takahashi S, Shin M, et al. MRI-based radiomics differentiates skull base chordoma and chondrosarcoma: a preliminary study. *Cancers (Basel)* 2022;14(13):3264.
20. Li L, Wang K, Ma X, et al. Radiomic analysis of multiparametric magnetic resonance imaging for differentiating skull base chordoma and chondrosarcoma. *Eur J Radiol* 2019;118:81–87.
21. Cilengir AH, Evrimler S, Serel TA, Uluc E, Tosun O. The diagnostic value of magnetic resonance imaging-based texture analysis in differentiating enchondroma and chondrosarcoma. *Skeletal Radiol* 2023;52(5):1039–1049.
22. Gitto S, Cuocolo R, Annovazzi A, et al. CT radiomics-based machine learning classification of atypical cartilaginous tumours and appendicular chondrosarcomas. *EBioMedicine* 2021;68:103407.
23. Gitto S, Bologna M, Corino VDA, et al. Diffusion-weighted MRI radiomics of spine bone tumors: feature stability and machine learning-based classification performance. *Radiol Med* 2022;127(5):518–525.
24. Zhang X, Su GH, Chen Y, Gu YJ, You C. Decoding intratumoral heterogeneity: clinical potential of habitat imaging based on radiomics. *Radiology* 2023;309(3):e232047.
25. Verma R, Correa R, Hill VB, et al. Tumor habitat-derived radiomic features at pretreatment MRI that are prognostic for progression-free survival in glioblastoma are associated with key morphologic attributes at histopathologic examination: a feasibility study. *Radiol Artif Intell* 2020;2(6):e190168.
26. Lee DH, Park JE, Kim N, et al. Tumor habitat analysis by magnetic resonance imaging distinguishes tumor progression from radiation necrosis in brain metastases after stereotactic radiosurgery. *Eur Radiol* 2022;32(1):497–507.
27. Lee DH, Park JE, Kim N, et al. Tumor habitat analysis using longitudinal physiological MRI to predict tumor recurrence after stereotactic radiosurgery for brain metastasis. *Korean J Radiol* 2023;24(3):235–246.
28. Qiao J, Wu H, Liu J, et al. Spectral analysis based on hemodynamic habitat imaging predicts isocitrate dehydrogenase status and prognosis in high-grade glioma. *World Neurosurg* 2023;175:e520–e530.
29. Ge W, Fan X, Zeng Y, Yang X, Zhou L, Zuo Z. Exploring habitats-based spatial distributions: improving predictions of lymphovascular invasion in invasive breast cancer. *Acad Radiol* 2024;31(11):4317–4328.
30. Zhang Y, Yang C, Qian X, Dai Y, Zeng M. Evaluate the microvascular invasion of hepatocellular carcinoma (≤ 5 cm) and recurrence free survival with gadoxetate disodium-enhanced MRI-based habitat imaging. *J Magn Reson Imaging* 2024;60(4):1664–1675.
31. Liu HF, Wang M, Lu YJ, et al. CEMRI-based quantification of intratumoral heterogeneity for predicting aggressive characteristics of hepatocellular carcinoma using habitat analysis: comparison and combination of deep learning. *Acad Radiol* 2024;31(6):2346–2355.
32. Zhang Y, Chen J, Yang C, Dai Y, Zeng M. Preoperative prediction of microvascular invasion in hepatocellular carcinoma using diffusion-weighted imaging-based habitat imaging. *Eur Radiol* 2024;34(5):3215–3225.
33. Clarke MJ, Mendel E, Vrionis FD. Primary spine tumors: diagnosis and treatment. *Cancer Control* 2014;21(2):114–123.
34. Roesch J, Cho JBC, Fahim DK, et al. Risk for surgical complications after previous stereotactic body radiotherapy of the spine. *Radiat Oncol* 2017;12(1):153.
35. Dosovitskiy A, Beyer L, Kolesnikov A, et al. An image is worth 16x16 words: transformers for image recognition at scale. *arXiv* 2020. Preprint posted online October 22, 2020; doi:10.48550/arxiv.2010.11929.
36. Selvaraju RR, Cogswell M, Das A, Vedantam R, Parikh D, Batra D. Grad-CAM: visual explanations from deep networks via gradient-based localization. 2017 IEEE International Conference on Computer Vision (ICCV), 2017; 618–626.